REASEARCH PROJECT RISK ASSESSMENT FORM

SCHOOL OF SCIENCE AND ENGINEERING

August 2025, V5

This form must be completed prior to the commencement of work to ensure that a suitable and sufficient assessment of reasonably foreseeable risks to health and safety is made, and appropriate reasonably practicable controls identified and implemented.

Individuals working under this risk assessment have a legal responsibility to ensure they follow the control measures stipulated to safeguard the health and safety of themselves and others.

The guidance document for completing this form should be consulted here

SECTION 1

OUTLINE THE PROJECT (complete well in advance of the project)

Title of the project: Evaluating Adversarial Robustness in Medical Image Classification
Department: Computing
Location(s) of work (including fieldwork): Computer laboratory, personal laptop
Description of the project, including methods, fieldwork locations & duration: This project involves evaluating the robustness of deep learning models for medical image classification against adversarial attacks. Using publicly available medical image datasets (e.g., chest X-rays), the study will implement simple adversarial attack methods such as FGSM or PGD using Python (PyTorch). The evaluation will compare model accuracy on original versus perturbed images. The project is expected to run over approximately 7 months.
Serial No.: Per-47

SUPERVISOR RESPONSIBLE FOR MANAGING THIS PROJECT

Name: Dr. Perley Xu	Position: Lecturer
Department: Computing	Date: 2025/10/16

PERSON UNDERTAKING THIS PROJECT

Name: Ruifeng Yan	Signature: Ruifeng Yan

Date assessment undertaken: 2025/10/16	Position: Bsc student

SECTION 2

PERSONS AT RISK

Identify all categories of individuals who may be affected either directly or indirectly through the work activity and status.

Who is at risk?		Status
Staff	☒	Trained ☒ Competent ☐ Inexperienced ☐ Disability ☐ Supervision required ☐ Lone working ☐
Students	☒	Trained ☒ Competent ☒ Inexperienced ☐ Disability ☐ Supervision required ☐ Lone working ☒
Visitors	☐	Trained ☐ Competent ☐ Inexperienced ☐ Disability ☐ Supervision required ☐ Lone working ☐
Contractors	☐	Trained ☐ Competent ☐ Inexperienced ☐ Disability ☐ Supervision required ☐ Lone working ☐
Public	☐	Trained ☐ Competent ☐ Inexperienced ☐ Disability ☐ Supervision required ☐ Lone working ☐

PROTECTIVE EQUIPMENT REQUIRED

PPE is a last resort and engineering controls (fume cupboard or Local Exhaust Ventilation) should always be used in preference. Consult with your supervisor regarding Personal Protective Equipment required. Insert any EN standard or details such as FFP2, Versaflo RPE, gloves with break through time etc., steel toe capped boots to be worn at all times below. You can use the Chinese equivalent standards if preferred for example: Chinese national product standard ***GB 24541-2022 – Hand protection – Protective gloves against mechanical risks****. This requires wearing Gloves with break through time; another standard* ***GB 21148-2020-Foot Protection-Safety footwea****r, which requires wearing steel-toe-capped boots; for the mask, FFP2 in the EU standard is equivalent to the Chinese standard KN95, as specified in Standard* ***GB 2626-2019*** *Respiratory Protection - Non-Powered Air-Purifying Particle Respirator.*

Engineering control/s	N/A
PPE required (head, eye, ear, mask type, glove etc. Training requirements?	N/A

ENVIRONMENTAL RISKS

Possible risks to the environment either from the project or as an accidental release to the environment

N/A

WASTE

Is waste generated in this project: Yes ☐ No ☒	Chemical Waste Form

SECTION 3

IDENTIFICATION OF PROJECT HAZARDS & EVALUATION OF RISKS

Likelihood harm will occur

1. **Rare** - hasn't happened but could / once every 10 years
2. **Unlikely** - could happen / known to happen / once a year
3. **Possible** – could easily happen / once a month
4. **Likely** – could happen / once a week
5. **Almost certain** – occurs several times a day / daily

Severity of harm

1. **No injury, damage**
2. **Minor injury, first aid required**
3. **Injury requires doctor or hospital attendance / absence**
4. **Major injury resulting in permanent damage**
5. **Fatality**

Risk Level Matrix

	S1	**S2**	**S3**	**S4**	**S5**
L1	1	2	3	4	5
L2	2	4	6	8	10
L3	3	6	9	12	15
L4	4	8	12	16	20
L5	5	10	15	20	25

Risk Level	Risk Evaluation
1 to 5	**LOW** - No Further action necessary, but if something can easily be done to reduce furthe then do. Monitor all existing controls.
6 to 9	**MEDIUM** – Further controls must be considered, and situation must be monitored.
10 to 25	**HIGH –** STOP Activity then make immediate improvements- Contact Safety Advisor.

From your proposal above consider main hazards below: (including chemicals)	From your persons at risk (1.3) what actions might cause harm? Think about HOW harm might occur. (Not injury)	List any existing controls in place now.	Risk level (L x S)	Identify any additional controls required (including supervision and training). Cross reference with action register below	New risk level (L x S)
Electrical - Coding on computer and using projector and camera	Student undertaking project – Electric shock	Check cables are in good condition;	L1 x S1 = 1		
Heat – generated by computer	Student undertaking project - can get hot.	Allow equipment to cool before moving/handling	L2 x S1 = 2		
Tight deadlines for project given work required for other modules	Lack of sleep and mental anguish	Project plan in place to ensure not crunching for deadlines.	L3 x S1 = 3		
Display screen equipment, non-ergonomic workspace.	Student undertaking project - Repetitive strain/Desk based/Eye strain	Ensure working environment has good ambient light; Take frequent breaks Refer to the University Display Screen Equipment Policy	L1 x S1 = 1		

From your proposal above consider main hazards below: (including chemicals)	From your persons at risk (1.3) what actions might cause harm? Think about HOW harm might occur. (Not injury)	List any existing controls in place now.	Risk level (L x S)	Identify any additional controls required (including supervision and training). Cross reference with action register below	New risk level (L x S)

Action Register

Action register List all additional controls required	Person responsible for implementing	Supervisor responsible	Timescale	Completed
None required				

SECTION 4

SUMMARY OF SIGNIFICANT FINDINGS

Please summarise; an evaluation of the risks involved; the precautions that are in place (or should be taken) and identify any actions which would improve the control of the identified risks. State whether there is a requirement for a written COSHH form and/or a Standard operating procedure to ensure correct safety measures are followed.

Insert links to relevant COSHH or Risk assessments:	None required
Insert links to relevant SOP	None required
Summary of assessment and required actions: Click here to enter text.	This project presents a very low overall risk. All activities are computer-based, involving programming, model training, and data analysis using anonymised public medical image datasets. There are no laboratory, chemical, or fieldwork components. The main potential risks include eyestrain and fatigue from prolonged computer use, data loss, and overheating of computing equipment during model training. These have been effectively controlled through ergonomic workstation setup, regular breaks, data backup routines, and proper ventilation. All data used in this project will remain anonymised and stored securely. No sensitive or personal medical information will be processed. No further

	COSHH or additional safety documentation is required.

SECTION 5

AGREEMENT TO THE TERMS OF THE PROJECT RISK ASSESSMENT

I confirm that, to the best of my knowledge, this project risk assessment describes all the significant risks associated with this project as well as any appropriate controls to reduce the risk to a level as low as is reasonably practicable. I agree to review this project assessment if there is an incident or changes to the activity/ies described above:

Title	Name	Signature	Date
Project Supervisor	Perley Xu	Perley Xu	16/10/2025
Student	Ruifeng Yan	Ruifeng Yan	16/10/2025